Do Now · retrieval

Lesson 060 · Refining an Abstraction and Testing Assumptions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does abstraction remove?

Q2

What is a model assumption?

Q3

What is a boundary test?

Q4

Two rising-edge D-type flip-flops A and B share a clock and start with QA=0 and QB=0. DA=1 throughout, and DB is connected to QA. Explain why QB is still 0 immediately after the first rising edge. Assume ideal simultaneous sampling.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

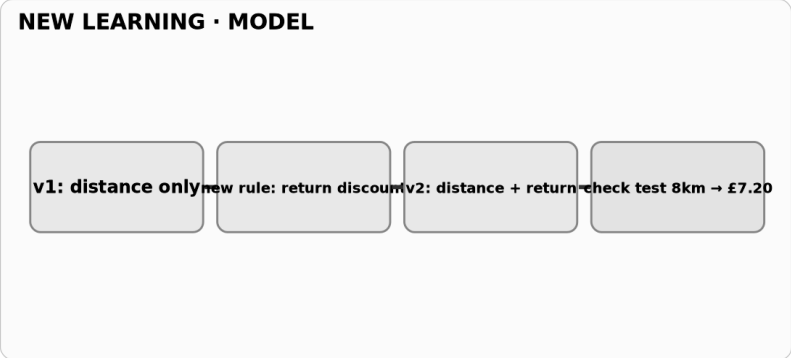
Core idea 2

Model — Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

Core idea 3

Evaluate — Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

refinement — Improving model for new requirements.

boundary — Important threshold or edge.

validity — Whether model works for its intended use.

assumption — Rule taken as given.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

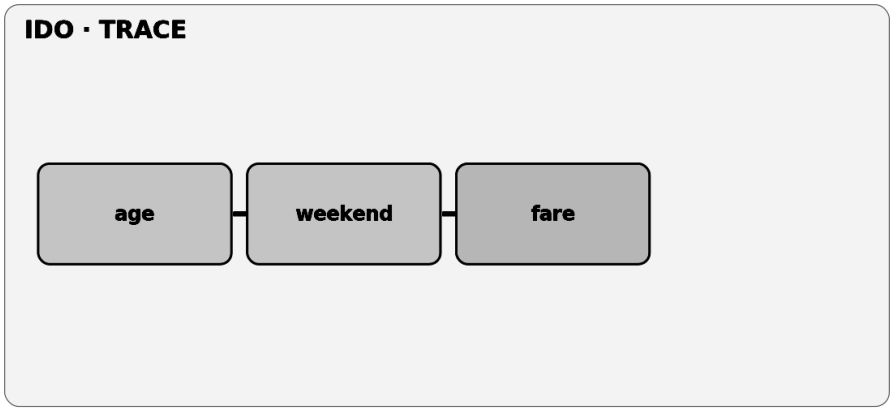
Problem

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

Follow the worked method

1. Explain — Weekend becomes a relevant distinction because it can change the output.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Cap is new relevant rule. Example 5 hours costs £12, not £15.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Ticket v1 charges £2 at ≤5km, £4 otherwise. Give charge at 5km and 6km and why these are useful tests. [6]

Question 2

Return journeys cost 1.8 times single. For 8km journey calculate return fare and show variables. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A school model predicts homework concerns from just missed work counts; discuss which relevant conditions to add and how to test impact. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A model that worked yesterday never needs changing when requirements change.

Use key vocabulary from memory

1. Explain refinement in your own words.

2. Explain boundary in your own words.

3. Explain validity in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.